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10 CONCEPTUAL Q&A NETWORKING · PRACTICE WORKSHEET

CONCEPTUAL SERIES

Service Mesh

Name: ______________________ Cohort/Batch: ______________________ Date: ______________________ Score: ______

SERVICE MESH PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Networking

TOTAL: 10 QUESTIONS

Q1 What is Service Mesh and what problem in Networking does it solve? *Threat Model*

Q2 What are the core components or concepts in Service Mesh? *Architecture*

Q3 How does Service Mesh compare to its main alternatives? *Configuration*

Q4 How do you get started with Service Mesh for a new project? *Hardening*

Q5 What configuration options most affect Service Mesh's behavior in production? *SSO / IdP*

Q6 How does Service Mesh handle reliability and high availability? *Observability*

Q7 What observability does Service Mesh provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Service Mesh? *Compliance*

Q9 What security best practices apply when running Service Mesh? *Testing*

Q10 What mistakes do teams commonly make when adopting Service Mesh? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Service Mesh is a tool or platform — Mitigation

Service Mesh is a tool or platform that automates or simplifies a specific concern in the Networking domain, reducing manual effort and operational complexity for engineering teams.

A2 Service Mesh has key building blocks that — Primitives

Service Mesh has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Service Mesh trades off simplicity against feature — Hardening

Service Mesh trades off simplicity against feature richness differently than its alternatives. Choose Service Mesh when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Service Mesh via its package manager — Gotchas

Install Service Mesh via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency — Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Service Mesh provides HA through leader election, — Observability

Service Mesh provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Service Mesh exposes metrics (Prometheus endpoint or — Automation

Service Mesh exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured — Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Service Mesh with the principle of — Assessment

Run Service Mesh with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and — Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.